Project Assignment
SC5010/CE9010 Introduction to Data Analysis

Important Contact:

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Due Date:

- Group Formation : before Week 3
- Project Presentation : Week 13: Monday and Wednesday
- Report Submission : before End of Week 13 (Sunday Midnight)

I. Introduction

This project aims at familiarizing the concepts of data analysis learnt through the course to provide some insights into the topic of interest. With the available software such as python; and analyzing the datasets obtained from public databases.

In this assignment, you will have the opportunity to apply the concepts and techniques you have learned in the course to a real-world dataset. You will choose a dataset from the four given datasets, and you will use software tools like Python to analyze the dataset, uncovering data characteristics, trends, or patterns, and compile a comprehensive report on your findings.

Another aim is to promote teamwork and self-learning: Working in a team pays big dividends, it is less stressful and peer support does make learning much easier. Furthermore, in line with NTU's vision to 'Teach less, learn more', you are encouraged to take the basic skills and principles, coupled with self-reading to handle real work problems.

II. GENERAL GUIDELINES:

- This project is a required part of the course. It shall account for the main coursework component in your final grade.
- This project is to be accomplished in a ***group of a Maximum four (4) students***. A supportive and conducive environment within each group is beneficial; hence you are free to determine your group members. While the accomplishments must genuinely belong to your group, you are free to have discussions with me and my teaching assistant, and most importantly, your classmates. The utilization of the ***discussion board*** in the NTULEARN is strongly encouraged. Bonus Marks will be awarded to students who have taken a contributing role in the discussion of this course, who has been interactive and has demonstrated generosity in assisting

others in understanding. This is a special thanks to these outstanding individuals and an encouragement to others.

III. Project Topics for Data Analysis

Project Objective

The students are expected to practice hands-on skills for how to perform a real-world Data Analytic task from the beginning (data pre-processing - data collection, cleaning, etc) to the final stage (data post-processing - evaluation and presentation, etc).

You are encouraged to test your approach using existing tools and implementations. After familiarizing the basic steps and seeing the results, you are encouraged to write your codes and submit them together with your report.

Before You Start

Plenty of tools are available for data analytics tools and platforms are freely available for data analysis. Simply searching the internet will return many lists of popular open-source data mining tools available.

Pay attention to the limitations of these tools and platforms in handling datasets. These will affect your selection and increase the challenge and complexity: Algorithms on a large dataset will run very slowly. And the tasks on a very small dataset may not be reflective of your capability.

IV. Selected Datasets for Your Project

1.Wine Quality Dataset

Official URL: https://archive.ics.uci.edu/dataset/186/wine+quality
Suggested Tasks: predict wine quality, classify wines, or find correlations between variables and quality and so on.

2. Heart Disease UCI Dataset

Official URL: https://archive.ics.uci.edu/dataset/45/heart+disease
Suggested Tasks: develop predictive models, analyze risk factors, or compare classification algorithms and so on.

3. Diabetes Dataset

Official URL: https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database
Suggested Tasks: predict diabetes onset, identify key predictors, or perform risk analysis and so on.

4. Google Play Store Apps Dataset

Official URL: https://www.kaggle.com/datasets/lava18/google-play-store-apps
Suggested Tasks: Analyze app popularity, predict success factors, or categorize apps and so on.

Your task is to select one of the datasets and conduct a thorough analysis. The topic is open-ended but must be relevant to the course themes.

V. Suggested Approaches

When developing your analysis, it is important to follow good approaches to the problem. This may include properly defining the problem, identifying relevant variables and data sources, developing a plan for data cleaning and preparation, and clearly stating your research questions or hypotheses. It is also important to prototype your analysis, by working through preliminary versions of your code and visualization to ensure that they are correct and that you are on the right track.

What do you do when you have a large dataset?

Performing sampling to select a subset of the data can reduce the data size to be analyzed. Obtaining the entire set of data of interest is too expensive and its analysis is time-consuming. And if you find out your approach does not work, you have not wasted too much time.

VI. Marking Scheme

This is a rough guide on how a marking will be done.

Assignment Grading Criteria

- **Technical Depth**
  How challenging is your selected problem? How difficult is your selected methodology/solution? Is it trivial to implement the selected idea? What kinds of tools/knowledge/codes are required to implement the chosen approach?

- **The significance of Experimental Results**
  Are your experimental results significant? Can your results answer the question or achieve the objectives of your application? What insight you have inferred?

- **Project Report**
  This is to evaluate the quality of your project report, including the organization, presentation, and comprehensiveness of the write-up.

- **Code Execution**
  How well does the submitted code run? Does it execute without errors? Does it efficiently achieve the desired outcomes? Consider the reproducibility of the results when the code is executed on different machines or environments

VII. What should be included in your project report?

Cover page: your group ID, your team members and their student ID (and their respective contribution in %) (The template of cover page will be provided)

- **Abstract**
  Summarize your whole project
- **Problem Description**
  - Motivation
  - Problem Definition
  - Related Work
- **Approach**
  - Methodology
  - Algorithms
- **Implementations**
- **Experimental Results and Analysis**
  - Experimental Setup
  - Comparison Schemes
  - Results and Analysis
- **Discussion of Props and Cons**
- **Conclusions**
  - Summary of project achievements
  - Directions for improvements
- **References**
- **Appendix:**
  - **Source Codes**

VIII. Submission Guidelines

For late submissions, a penalty of **1 mark** per day will be applied after the deadline. The assignment will not be accepted if more than a **7-day delay**. Please remember to submit your assignment before the deadline.

What To Submit:

1. A file called **Project_Report_Group_XX.docx.** Please show your group members' names and IDs on the cover page of your project report. If you are using Latex, submit the latex source and the compiled PDF.
2. A README file. Please name it **README.txt** This file should include three sections:
    - Your group ID and group member names
    - Source code with detailed instructions and documentation on how to reproduce your results.
3. The page limit of the report is 20 pages. The over-length case may be penalized.

Submission Instructions

Submit the package file with the Subject “**PROJECT SUBMISSION GROUP** XXXXXXX”, where XXXXXXX in the email subject is your group leader name in uppercase, to the NTU learn course site and send an email to the respective TAs to keep them in the know.

Please package all of your files (including your report "Project_Report_Group_XX.docx" the README.txt file, scripts and source code) into a ZIP file, named it “Project_Group_XXXXXXX.zip”, where XXXXXXX is your group Leader name in upper case.